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Programming for Data Analytics

PDAN8411 Part 1

Technical Report

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Introduction

When it comes to the healthcare area and aspect, especially if we investigate the medical aid schemes, it is vital and extremely important to be able to display accurate medical expenditures. Predicting healthcare costs is an important element of risk management in the insurance space (McKinney,2010). It not only gives you a technological advantage, but it is also a very big factor in terms of having a maintainable insurance management. In this report, we are looking at the elements of linear regression, which is a very important element of supervised learning, on a data set containing demographic and health-related features. The main goal of this report is to formulate a framework (mathematical) that can forecast insurance charges based on several different independent variables.

When we look at the different prices and how they arise in healthcare, different factors get taken into consideration. When we investigate the data analytics, we do not focus on the normal and market structure of the medical aim scheme pricing, but we focus on personal capabilities. Within this data analytics project, we are going to investigate data preprocessing, theoretical modeling, evaluation, and the interpretation of statistical metrics within a business context.

The Mechanics of Linear Regression

When we investigate the definition of linear regression, it is a statistical method/model that looks at the relationship between a dependent variable and one or more independent variables by putting it in a linear equation to analyze the data. This looks at the relationship between a scalar response and one or more explanatory variables (Montgomery, Peck, and Vinning, 2021). Now, when we investigate our analysis and investigate our project, the dependent variable is the variable of Y, and the independent variables are the variables of age, BMI, and smoker status.

The Mathematical Equation

Below is the model that follows linear regression:

$$Y = Bo + B1X1 + B2X2 + BnXn + E$$

Where:

Y= looks at the healthcare charges

Bo(Intercept)= Baseline costs where all predictors are zero (investigates the theoretical baseline)

B1 Bn (Coefficient) = The weight assigned to each feature, indicating the change in Y for every unit change in X

E(Error Term)= Represents the variance that cannot be explained by the independent variables

Statistical Assumption

If we want the linear regression to be unbiased, fair and valid, assumptions need to be made:

Linearity = This looks at the relationship between independent and dependent variables, which must be very additive and linear.

Independence = Observations must be independent of each other

Homoscedasticity = the errors within the machine must be constant throughout all the independent variables.

Normality = a normal distribution should be followed

When we investigate this analysis, I have used the Ordinary Least Squares, and what that does is that it squares certain points so that all the points are the smallest that they can possibly be when it comes to getting them as close as possible to the line within linear regression. It focuses on the smallest number.

Data Processing and Feature Engineering

When we investigate the integrity of any machine, it depends on the value of the data that it takes in. Mathematical models require numerical matrices to perform (Pedregosa et al.,2011). If we investigate raw data, that is, unstructured data and skewed variables, then that impacts the reliability of the data.

Categorical Encoding

Looking into machine learning, the only way it can come out a success and without any errors is if it uses numerical inputs, and when we investigate our data set, variables like "sex", "BMI", "Region" are categorical, and we classified them as follows in our project:

Label Encoding:

Label encoding is assigning categories into numerical values, and in our project, we are taking, for example, "sex" as the numerical value of 1, and "smoker" as 0. What this does is that in coding, it allocates categories and more words and theoretical elements into numerical values.

One-Hot Encoding:

What one-hot encoding does is that it takes variables and puts them into elements where they are not greater than one another. For example, it will take the regions, such as Northwest or Southeast, and make sure that these variables are not greater/higher than one another.

Data Splitting Strategy

If we do not want the data to be overdoing itself, or fail to capture new information, we use a dataset that has an 80/20 split ratio. The 80% training set allows the model to learn coefficients, while the 20% test set serves as an objective benchmark for performance evaluation.

Exploratory Data Analysis (EDA) Insights

An exploratory analysis was done before the modeling. When we look into observing the data, smokers have a higher medical charge than non-smokers, and if we are looking at it from a theoretical point of view, medical facilities view that you are intentionally putting your life (lungs) in danger, so that is why the charges will be higher than those of non-smokers. This suggests a high positive coefficient for this variable in the final model. Also, BMI showed a positive correlation, especially in people above the threshold for obesity (30.0).

Performance evaluation and interpretation

This project's performance was measured using two factors: the coefficient of Determination (R2) and the mean squared error (MSE)

R-Squared Analysis

The model achieved an R2 score of 0.8069. In data analytics/statistical terms, this means that 80.7% of the dependent variable (charges) is based on the independent variables (Age, BMI, Smoking status, etc). In terms of sciences, an R2 of 0.8 is a good mark, and it indicates a very strong fit for the model.

Error Metrics (MSE)

This looks at the average errors within the data. When the value can seem high due to the currency values, it is vital when it comes to the outliers. This is especially important in insurance when it comes to high-cost and low-cost patients.

Business Implications and Business Future Works

This element has a positive aspect to it because when it comes to the geographical area, this element can help with the instantly generated cost profile. So, it can automatically give you a cost profile for your geographical area.

We look at elements such as feature scaling, such as looking at the BMI and age, to see if the coverage is faster. We also investigate polynomial features, which look at the relationship between age and costs as quadratic rather than linear. Then there is regularization, which looks at features not dominating the model excessively.

Conclusion

This project and analysis successfully demonstrate the efficacy of Linear Regression in predicting medical costs. When we look at the result of 80.7%, it shows that this model is a reliable framework for data-driven decisions.

Reference list

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